

Examiner reports

Q1.

A considerable number of candidates could not factorise the quadratic expressions correctly and there were many poor attempts at division. It is possible that candidates are familiar with working with cubic expressions and had difficulty applying the same techniques to quadratic expressions.

Q2.

- (a) (i) It was good to see that almost all candidates started correctly by evaluating $p(2)$, though a few thought they needed to find $p(-2)$ and others wrongly assumed that long division was the “factor theorem”. It was necessary to write a conclusion or statement after showing that $p(2) = 0$, in order to earn the second mark.
- (ii) The most successful approach here was by using a quadratic factor ($ax^2 + bx + c$), though long division also worked well for many. A surprising number who found the correct quadratic then factorised it wrongly. Those who tried the factor theorem again rarely spotted both factors. A few lost the final mark by failing to write $p(x)$ as a product of factors.
- (b) Although there were many correct sketches, many lost a mark by failing to mark the point $(0,8)$ on the y -axis. Candidates were expected to draw a cubic through their intercepts, to use an approximately linear scale and to continue the graph beyond the intercepts on the x -axis. It was common to see the negative values in the factors wrongly taken to be the roots and hence the intercepts on the x -axis.

Q3.

This question was done well with nearly all candidates getting the correct value of 0 for parts (a) and (b), but in part (b) some candidates were penalised for not explicitly evaluating power of -1 and so not obtaining the given answer convincingly. Most candidates attained the correct answer for part (c) although some did considerably more work than others in gaining the marks. Some used the factors from parts (a) and (b) to write down the values of a and b by inspection whilst others multiplied the three linear factors and then equated coefficients. Others used division either dividing by the quadratic factor or one or both of the linear factors.

Q4.

Part (a) Although many candidates showed that $p(3) = 0$, many lost a mark for failing to include a statement of the implication. Some candidates appeared ignorant of the Factor Theorem and used long division and therefore earned no marks in this part.

Part (b) Only about half of the candidates were able to complete this part, although most made an attempt. The term $x^2 - x$ confused some. A few failed to write a **product of factors** even though this was requested.

Q5.

Part (a)(i) Most candidates correctly answered this; even those who did not usually went on to assume that $(x - 2)$ was a factor of the polynomial.

Part(a)(ii) Most candidates were successful here, but those who used division rather than the requested factor theorem gained no credit as they had not answered the

question. Similarly those candidates who just wrote $f\left(-\frac{1}{2}\right) = 0$ were penalised because without showing some arithmetic, they had not demonstrated the result; a conclusion interpreting the result of zero, was also required.

Part (a)(iii) the intention here was that candidates would use inspection, having found two factors in parts (i) and (ii). Although many did this, other candidates proceeded by division, either by a linear factor or the quadratic factor, and thus took more time than was necessary. Most candidates successfully factorised the polynomial, although methods, and thus the time taken, did vary considerably, some using $(ax + b)$ for the third factor, multiplying out and equating coefficients.

Part (b) Most candidates did this correctly, although some apparently could not factorise $3x^2 - 6x$ or went only as far as $3(x^2 - 2x)$, and either did some illicit cancelling or abandoned the attempt. A relatively large number of candidates interpreted “simplify” as meaning put the expression into partial fractions. Although this was accepted, it took a great deal of time-consuming algebraic manipulation to get the correct result, and few did so.

Q6.

- (a) (i) Some were confused by the word “factor” and gave answers as 1 and -2 .
- (ii) Although there were many correct answers given for the factorisation, some candidates appeared to be guessing the factors of $f(x)$, making no use of the information given in part (a)(ii). Despite getting a non-zero remainder for part (a)(i), several candidates gave $(x + 1)$ as a factor.
- (b) Most candidates found the coordinates of A , and those with the correct factor $(x - 4)$ of $f(x)$ were usually able to find the correct coordinates of B .
- (c) It was pleasing to see most candidates able to integrate correctly, although quite a few failed to integrate 8 to give $8x$. Some candidates did not evaluate the limits in the correct order, but there were many correct values seen for the area of the shaded region.

Q7.

- (a) It was disappointing to see a number of candidates unable to deal with the minus sign outside the second bracket. Another very common error was to simplify the expanded brackets correctly to $-11k^2 - 14k + 25$ and then to say that this was equal to $11k^2 + 14k - 25$, thus losing the accuracy mark.
- (b) (i) The sign in the printed answer caused many to write the condition for real roots as $b^2 - 4ac \leq 0$. There was a lot of ‘fudging’ in this part of the question and it was a rare event to see a candidate score full marks.

- (ii) The most successful candidates identified the critical values and then used a sign diagram or a sketch to identify the correct solution to the inequality.

Q8.

Use of the binomial expansion and use of the expansion of $(2 + x)(2 + x)^2$ were equally popular in part (a)(i) and most obtained the correct answer. There was however a minority of candidates who did not include the powers of 2 in the middle terms when applying the binomial expansion. It was disappointing to see so many candidates producing masses of algebra rather than replacing x by $(-x)$ to obtain the answer to part (a)(ii). There was some evidence of 'faking' to reach the printed answer in part (b), which quite often resulted in marks being lost as earlier correct work was changed. The vast majority realised that dy/dx was required in part (c), although some later explanations could have been more precise.

Q9.

This question was answered very well by many candidates, and evidently the basic processes of differentiation and integration had been well rehearsed.

Part (a)(i) Most candidates substituted $x = 3$ into the given cubic equation and found $p(x)$ to be zero. Some, however, lost a mark as they did not indicate that $p(3) = 0$ implied that $x - 3$ was a factor.

Part (a)(ii) The coordinates $(3, 0)$ were often seen but some forgot to give the y coordinate

Part (b)(i) Almost all candidates differentiated correctly, which was pleasing.

Part (b)(ii) Most candidates realised the need to equate dy/dx to zero. However not all were able to solve the resulting equation. Some of those who did solve the quadratic offered both solutions, 1 and $7/3$. Many found the second derivative in order to ascertain which point was the minimum point M . This was unnecessary as the coordinates of A $(1, 0)$ were clear from the diagram.

Part (c) The second derivative was almost always found correctly followed by the substitution of $x = 1$.

Part (d)(i) The integration was carried out very well. Occasionally $\frac{3x^4}{4}$ was seen but this was rare.

Part (d)(ii) Most candidates used the correct limits of 0 and 1. However many arithmetic errors occurred.

When trying to combine the fractions so that the correct value of the integral $-\frac{11}{12}$ was not seen very often. Of those who did integrate correctly, many still lost the last mark for not stating that the area of the shaded region was $11/12$ and the negative value of the integral arose because the region was below the x -axis. It was not acceptable to simply

write Area = $-\frac{11}{12} = \frac{11}{12}$.

Q10.

Part (a) Practically everyone made a reasonable attempt to multiply out the brackets but the minus signs caused problems for the weaker candidates, particularly when trying to combine like terms.

Part (b) Usually candidates were able to find the single real root 2. It was expected that candidates would have found the discriminant of the quadratic and, since this was negative, conclude that the quadratic equation had no real roots. All too often, candidates tried to find " $b^2 - 4ac$ " for the cubic $x^3 - x^2 + x - 6$.



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